

DES PUNE UNIVERSITY
School of Engineering and Technology
Computer Engineering and Technology
Program: B.Tech. Computer Science and Engineering

Academic Year: 2025-26

Year: Second Year

Term: I

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Subject: Java

Assignment No.: 2

Title: Student Management System

Date: 04/08/2025

Java
Assignment 2

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Aim: $\mathbb{R}$

Implement a class with multiple constructors to demonstrate constructor overloading, specifically within the framework of a student management system.

Theory:

1. Constructor

A constructor in java is a special method that is used to initialize objects. It is called when an object of a class is created.

Syntax:

```
class Student  
{
```

```
    Student()  
{
```

```
        // Default constructor  
    }
```

```
}
```

```
class Student  
{
```

```
    Student (String name, int age)  
{
```

```
        name = name;
```

```
        age = age;
```

```
        // Parameterized constructor  
    }
```

```
}
```

2. Copy constructor

We can create a copy constructor by passing an object of the same class as a parameter.

Syntax:

```
Student (Student S)
{
    name = S.name;
    age = S.age;
}
```

3. Inner class

It is a class defined within another class.

Syntax:

```
class Outer
{
    class Inner
    {
        void fn()
        {
            //Statement
        }
    }
}
```

4. Enumeration

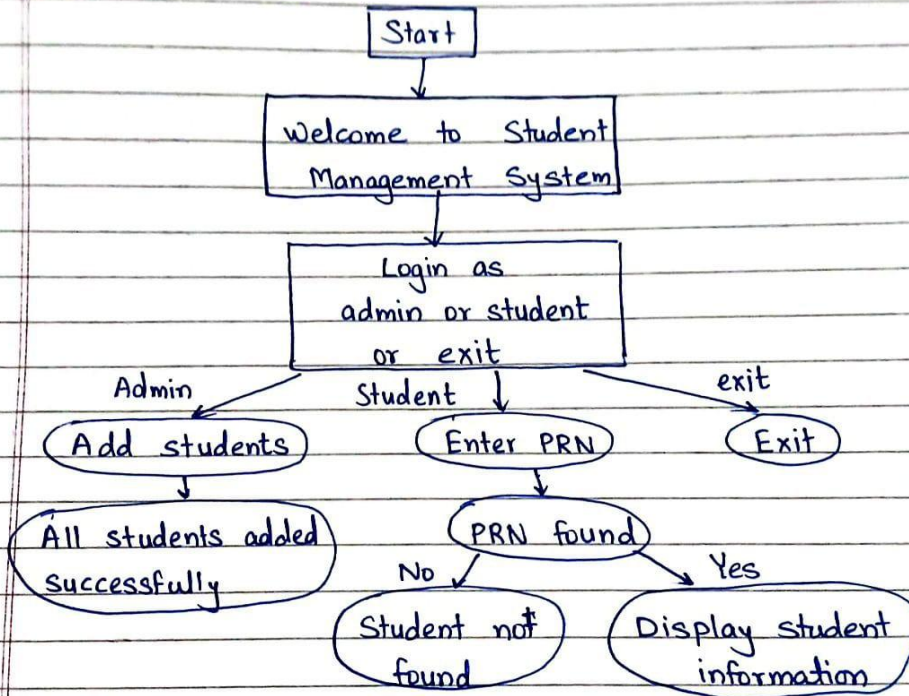
Enum is a special java type used to define a collection of constants.

eg. enum Days

```
{
    MON, TUES, WED, THURS, ...
}
```

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Flow chart:



Code:

```
package Assignment2;

import java.util.Scanner;

public class StudentManagement
{
    static Scanner sc = new Scanner(System.in);
    static StudentDatabase database = new StudentDatabase();
    static int totalStudents = 0;

    public static void main(String[] args)
    {
        String login;
        System.out.println("Welcome to Student Management System");

        do
        {
            System.out.println("\nLogin as (admin/student) or type exit to quit:");
            login=sc.nextLine();

            if(login.equals("admin"))
            {
                adminLogin();
            }
            else if(login.equals("student"))
            {
                studentLogin();
            }
        }
    }
}
```

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```
    }
    else if(!login.equals("exit"))
    {
        System.out.println("Invalid login type");
    }
}while (!login.equals("exit"));
System.out.println("Exiting Student Management System");
}

static class StudentDatabase
{
    static Student[] students= new Student[100];
}

enum Year
{
    FY,SY,TY;
}

public static void adminLogin()
{
    System.out.println("How many students do you want to add? ");
    int count = sc.nextInt();
    sc.nextLine();

    for(int i=0;i<count;i++)

    {
        System.out.println("\nEnter details for students" + (i+1) + ":");
        System.out.println("Name: ");
        String name= sc.nextLine();
        System.out.println("School: ");
        String school= sc.nextLine();
        System.out.println("Branch: ");
        String branch= sc.nextLine();
        System.out.print("Year(1 - FY, 2 - SY, 3 - TY): ");
        int yearNum= sc.nextInt();
        Year year=null;

        if(yearNum == 1)
        {
            year=Year.FY;
            StudentDatabase.students[totalStudents] = new Student(name, school, branch, year);
        }
        else if(yearNum == 2)
        {
            year= Year.SY;
            System.out.println("SGPA: ");
            double sgpa = sc.nextDouble();
            System.out.println("CGPA: ");
            double cgpa = sc.nextDouble();
            StudentDatabase.students[totalStudents] = new Student(name, school, branch, year, sgpa, cgpa);
        }
    }
}
```


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```
        else if(yearNum == 3)
        {
            year= Year.TY;
            System.out.println("SGPA: ");
            double sgpa = sc.nextDouble();
            System.out.println("CGPA: ");
            double cgpa = sc.nextDouble();
            StudentDatabase.students[totalStudents] = new Student(name, school, branch, year, sgpa, cgpa)
        }
        else
        {
            System.out.println("Invalid year. Skipping this student.");
            sc.nextLine();
            continue;
        }
        totalStudents++;
        sc.nextLine();
    }
    System.out.println("\nAll students added successfully.");
}

public static void studentLogin()
{
    System.out.println("Enter your PRN: ");
    int inputPRN = sc.nextInt();
    sc.nextLine();

    boolean found = false;

    for(int i=0;i< totalStudents; i++)
    {
        if(StudentDatabase.students[i].prn == inputPRN)
        {
            StudentDatabase.students[i].display();
            found=true;
            break;
        }
    }

    if(!found)
    {
        System.out.println("Student with PRN "+inputPRN+ " not found.");
    }
}
}

class Student
{
    static int prnCounter=1000;
    int prn;
    String name;
    String school;
    String branch;
    StudentManagement.Year year;
```

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```
double sgpa;
double cgpa;

// 1 use of this to refer instance variables
Student(String name, String school, String branch, StudentManagement.Year year)
{
    this.prn = prnCounter++;
    this.name = name;
    this.school = school;
    this.branch = branch;
    this.year = year;
    this.sgpa = 0.0;
    this.cgpa = 0.0;

    // 4 this passed as method argument
    printStudent(this);

    // 3 this used to call current class method
    this.displayPartial();
}

// 2 use of this() to call another constructor (constructor chaining)
Student(String name, String school, String branch, StudentManagement.Year year, double sgpa, double cgpa)
{
    this(name, school, branch, year); // calls first constructor
    this.sgpa = sgpa;
    this.cgpa = cgpa;
}

// Copy constructor
Student(Student other)
{
    this.prn = other.prn;
    this.name = other.name;
    this.school = other.school;
    this.branch = other.branch;
    this.year = other.year;
    this.sgpa = other.sgpa;
    this.cgpa = other.cgpa;
}

// 6 this returned from method
Student getStudent()
{
    return this;
}

// 4 this passed as method argument
void printStudent(Student s)
{
    System.out.println("Welcome, " + s.name);
}

// 3 this to call current class method
void displayPartial()
```

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```
{
    System.out.println("Student " + this.name + " added successfully");
}

void display()
{
    System.out.println("\n~ Student Information ~");
    System.out.println("PRN: " + this.prn);
    System.out.println("Name: " + this.name);
    System.out.println("School: " + this.school);
    System.out.println("Branch: " + this.branch);
    System.out.println("Year: " + this.year);
    if(this.year == StudentManagement.Year.FY)
    {
        System.out.println("Branch is visible for FY students");
    }
    else
    {
        System.out.println("SGPA: " + this.sgpa);
        System.out.println("CGPA: " + this.cgpa);
    }
}

class Activity
{
    void displayActivity()
    {
        {
            System.out.println("Other activities: Sports, Cultural");
        }
    }
}

// 5 this passed as constructor argument to another class
void callHelper()
{
    Helper h = new Helper(this);
    h.showStudent();
}
}

class Helper
{
    Student s;
    Helper(Student s)
    {
        this.s = s;
    }
    void showStudent()
    {
        System.out.println("Displaying student via Helper class");
        s.display();
    }
}
```

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Output:

```
Welcome to Student Management System

Login as (admin/student) or type exit to quit:
admin
How many students do you want to add?
2

Enter details for students1:
Name:
Riddhee
School:
Engineering and Technology
Branch:
CSE
Year(1 - FY, 2 - SY, 3 - TY): 2
SGPA:
9
CGPA:
9.1
Welcome, Riddhee
Student Riddhee added successfully

Enter details for students2:
Name:
Nandini
School:

Business and Management
Branch:
FinTech
Year(1 - FY, 2 - SY, 3 - TY): 1
Welcome, Nandini
Student Nandini added successfully

All students added successfully.

Login as (admin/student) or type exit to quit:
student
Enter your PRN:
1000

~ Student Information ~
PRN: 1000
Name: Riddhee
School: Engineering and Technology
Branch: CSE
Year: SY
SGPA: 9.0
CGPA: 9.1

Login as (admin/student) or type exit to quit:
exit
Exiting Student Management System
```

Conclusion:
This program demonstrates the concept of constructor overloading, enum usage and java inner classes